

Ishwor Raj Pokharel

Suryabinayak-2, Bhaktapur, Nepal

☎ (+977) 9845964936 | ✉ pokhrel.ishwor13@gmail.com | 📄 Ishwor-git

Summary

Final-year Electronics, Communication and Information Engineering student with hands-on experience across deep learning architectures spanning medical imaging, speech processing, and edge AI. Co-author on a paper accepted for publication on multimodal audio-visual speech extraction (IsoNet), with additional work in graph neural networks, transformers, and LLM hallucination detection.

Work Experience

Cloudfactory Nepal

Kathmandu, Nepal (Remote)

DATA ANNOTATOR

Nov. 2022 - Sep. 2023

- Developed high-quality datasets for CV models by performing precise image labeling, bounding box creation, and complex polygon segmentation.
- Curated labeled datasets from unstructured documents to optimize Information Extraction models.
- Conducted rigorous evaluation of model predictions to identify edge cases
- Worked remotely coordinating with colleagues from various countries

Education

Institute of Engineering, Tribhuban University

Kathmandu, Nepal

B.E. IN ELECTRONICS COMMUNICATION AND INFORMATION ENGINEERING

Mar. 2023 - Present

- Relevant Coursework: Discrete Structures, Data Structures and Algorithms, Database Management Systems, Probability and Statistics, Computer Networks, Communication Systems, Wireless Communications, Digital Signal Processing, Data Mining (Elective), Artificial Intelligence
- Projects : Facial Expression Detection using opencv(cpp), Web based Exam Organization and proctoring system, Analysis of Associative Rule Mining in Microarray dataset

Microdegree and Certification

Fusemachines

Kathmandu, Nepal

MICRODEGREE IN ARTIFICIAL INTELLIGENCE 2025 (AI FELLOWSHIP)

Mar. 2025 - Dec. 2025

- Grade: 76.52%
- Core concepts learnt : Tree based models Deep Neural Network, Image Processing deep computer vision, Natural Language Processing, Sequential Models, Transformers, Foundational

Nepal Applied Mathematics and Informatics Institute for Research (NAAMII)

Kathmandu, Nepal

6TH ANNUAL NEPAL AI SCHOOL 2026

Dec. 2025 - Jan. 2026

- Completed a 10-day intensive AI program (46 hrs lectures, 12 hrs labs, 4 hrs Ideathon) covering geometric deep learning, graph neural networks, transformers, reinforcement learning, and LLM foundations.
- Participated in the Ideathon, pitching a group project on Digital Twin-Driven Microgrid Optimization.
- Engaged with topics including AI ethics, AI in healthcare, hallucination detection in LLMs, and trustworthy graph ML through hands-on lab sessions.

Projects

ISO-Net : Multimodal Audio-Visual Target /speech Extraction

HACKATHON WINNING PROJECT

- Built a multimodal system to solve the cocktail-party problem, isolating a user-selected speaker's voice from noisy multi-speaker audio using a 4-microphone array.
- Combined spatial audio cues (GCC-PHAT) with face-conditioned visual embeddings inside a U-Net mask estimation network to enable face-selectable speech extraction.
- Achieved 9.31 dB SI-SDR (a 4.85 dB improvement over the raw mixture) on a hard -1 to 10 dB SNR test set, outperforming oracle delay-and-sum and MVDR beamformers.
- Won 1st place at KU Hackfeast 2025 and co-authored a resulting paper accepted for publication in JIEE.

Human Presence Detection System Using Distributed Inference on Resource-Constrained Edge Devices

THIRD YEAR PROJECT

- Designed a distributed, tile-based computer vision system to enable real-time human presence detection on low-power microcontrollers.
- Implemented adaptive image tiling on an ESP32-CAM using Sobel-filter edge density to prioritize informative regions before transmission.
- Deployed a Local Binary CNN based model with 82% accuracy on ESP32-S3 nodes, communicating over ESP-NOW for distributed inference.
- Aggregated distributed inference results on a master node to drive real-time detection status output via I2C LCD.
- Code : Github

Detection of Hallucinations in Output of Reasoning Chains by LLMs

AI FELLOWSHIP GRADUATION PROJECT

- Built lightweight classifiers to detect hallucinations in step-by-step reasoning chains generated by LLMs such as GPT.
- Benchmarked multiple architectures, from embedding-similarity (50 to 60% accuracy) baselines to an LSTM model with an attention layer (70 to 80% accuracy) , for hallucination detection.
- Curated custom dataset composed of various reasoning steps to solve language and logical problems that was used to train and evaluate
- code : Github

Spatiotemporal Graph Transformer for Clinically Interpretable Early Alzheimer's Disease

Detection Using rs-fMRI

ONGOING FINAL YEAR PROJECT

- Designing a spatiotemporal graph transformer to classify Alzheimer's Disease, Mild Cognitive Impairment, and Normal Controls from resting-state fMRI data in the ADNI dataset
- Combined Graph Attention Networks with a Temporal Transformer to jointly model spatial brain connectivity and its evolution over time.
- Applied self-supervised contrastive pretraining (InfoNCE loss) to improve robustness on limited clinical neuroimaging samples.
- Building interpretability into the pipeline using attention analysis and Integrated Gradients, identifying the hippocampus and Default Mode Network as key biomarkers.

Publications

IsoNet: Multimodal Audio-Visual Target Speech Extraction with Spatial Conditioning for Compact Microphone Arrays

*Journal of Innovation in
Engineering Innovation (JIEE)*

CO-AUTHOR

Accepted for publication

- Designed a multimodal target speech extraction system combining complex STFT features, GCC-PHAT spatial cues, and face-conditioned visual embeddings within a U-Net mask estimation network for 4-microphone arrays.
- Achieved 9.31 dB SI-SDR (a 4.85 dB improvement over the mixture) on a hard -1 to 10 dB SNR test set, outperforming oracle delay-and-sum and MVDR beamformers, which degraded performance by 4.82 dB and 6.08 dB respectively.
- Demonstrated consistent gains from visual conditioning and GCC-PHAT features via ablation studies, establishing a compact-array, face-selectable speech extraction baseline.

Technical skills

Programming Languages : C, C++, Python, Javascript, X86 assembly, Bash

Database : SQLite, PostgreSQL, MySQL, MongoDB

Framework and Libraries : FastAPI, Django, React, Numpy, Pandas, Pytorch

Embedded Hardware : Arduino, Raspberry Pi, ESP32

Tools : Docker, Git, Linux, Vim, VSCode, Jupyter Notebook, Google Collab, Conda

Languages: Nepali (Native), English (Fluent)

Extracurricular Activity

Tensor IOE

IOE Thapathali Campus

GENERAL MEMBER & PRESIDENT AT 2024

Nov. 2023 - Mar. 2026

- Run the society focusing on AI and Machine Learning and building network on and off campus.
- Organized various workshops, Webinars, Instructor sessions and competitive programs in the field
- Instructed numerous workshop on Machine Learning

KU Hackfeast 2025

*Kathmandu University, Dhulikhel,
Nepal*

1ST POSITION WINNER

Dec 2025

- Won 1st place for IsoNet, a multimodal target speech extraction system (see Projects).